

# Anim Assist

User Manual — Version 12.0.1 (low-jitter build)

Production animation workflow suite for **Bforartists**

Eleven tool phases · 400+ operators · built-in explainer help on every control

June 2026 · Bforartists 4.2+ exclusive

# 1. Overview

Anim Assist is a production animation suite organized into eleven feature phases, each living in its own sidebar tab so the toolset scales without burying you. It covers the full character-animation loop: key editing, breakdowns, trajectory polish, retiming, proxies, IK/FK matching, mirroring, layers, macros and lipsync.

Installation: *Preferences* → *Add-ons* → *Install from Disk* → `b4_anim_assist_v12.0.1_low_jitter.zip`. Bforartists 4.2+ only — on standard Blender the addon refuses to register and points to `bforartists.de`. After enabling, run **First Run Setup** (Workspace tab) to configure defaults, and optionally **Load Demo Config** to explore with sensible presets.

## Where everything lives

| Sidebar tab | Phase          | Contents                                                                       |
|-------------|----------------|--------------------------------------------------------------------------------|
| Keys        | Phase 2        | Key/channel editing power tools (also panels inside Graph Editor & Dope Sheet) |
| Pose        | Phases 3, 4, 9 | Breakdowns & inbetweens · modal transform offset · mirroring & symmetry        |
| Motion      | Phases 5, 6    | Trajectory overlay & arc diagnostics · retiming & timing diagnostics           |
| Rig         | Phases 7, 8    | Proxy creation & baking · IK/FK match & space switching                        |
| Workspace   | Phase 10       | Tool shelf, macros, audit trail, recovery, first-run setup, hotkey checks      |
| Layers      | Phase 11       | Animation layer stack & blending                                               |
| Lipsync     | Phase 12       | Hybrid PREVIEW/SHIPPED lipsync with shape keys, viseme libraries, cue tables   |

Built-in help: nearly every control carries a (?) **explainer icon** — over 400 curated entries — plus a searchable Help Browser. Toggle explainer icons (or icon-only mode) in Preferences. Pie menus: **Ctrl+Shift+D** and **Ctrl+Shift+K**. Header toolbars add one-click access in animation editors.

## 2. Keys — selection & channel power tools (Phase 2)

Fast key selection by range, playback/preview range, visibility, first/last, between-selected, inversion; channel isolation with push/pop stack (drill into a control, then climb back out); channel visibility management; key metadata: protect selected keys from accidental edits, prune orphan metadata, clear all; copy keys and paste at frame with offset handling.

Panels appear in the 3D View Keys tab and inside the Graph Editor / Dope Sheet sidebars so the tools are where your cursor already is.

## 3. Pose — breakdowns, offset, mirroring (Phases 3, 4, 9)

**Breakdowns & inbetweens (P3):** create breakdown poses biased toward either neighbor key, generate inbetweens, and compare poses side-by-side. Favoring sliders give classic 'thirds' workflows in one drag.

**Transform offset (P4):** a modal drag that offsets the current pose or whole motion with adjustable sensitivity (autokey-aware, with margins handling), letting you nudge animation without opening curve editors.

**Mirroring & symmetry (P9):** automatic left/right pair detection from bone names (49 unit tests back this matcher), pose mirroring across pairs, batch mirror with reporting, symmetry audits.

## 4. Motion — trajectories & retiming (Phases 5, 6)

**Trajectory overlay (P5):** draws editable motion paths for selected targets right in the viewport: position samples, keyframe ticks, velocity coloring. Eight arc-quality detectors (drift, flattening, zigzag, pops, spacing highs/lows, reversals, dead stops, apex contact) flag issues; navigation ops jump the playhead to each flagged frame.

**Retiming & timing diagnostics (P6):** retiming handles in Dope Sheet/Graph Editor plus a timing diagnostics pass (Run Diagnostics) that audits key spacing and rhythm.

v12.0.1 'low-jitter' defaults: the overlay is tuned to stay on while you animate — max samples 160, 3 simultaneous targets, frame ticks off, and **constraint evaluation is skipped during playback** (it is accurate but slow; enable it for IK-driven checks while paused). Arc detectors now run ONLY from the Run Diagnostics button — turning the overlay on no longer stalls your viewport.

## 5. Rig — proxies & IK/FK matching (Phases 7, 8)

**Proxies (P7):** generate lightweight proxy objects for heavy rigs, animate against them, then bake back to the real rig (31 math tests + 22 session tests ship with the addon).

**IK/FK match & space switching (P8):** snap IK to FK (and back) with axis-mask control over location/rotation/scale; switch spaces with compensation so the pose never jumps; switch-history navigation, contact preservation, opposite-side matching, batch operations and quick-action buttons.

## 6. Workspace — macros, audit & recovery (Phase 10)

A customizable tool shelf; a macro engine that chains Anim Assist operators into one-click sequences (including diagnostics passes); an audit trail of recent tool actions; recovery utilities.

System hygiene lives here too: **First Run Setup**, **Load Demo Config**, **Check Hotkey Conflicts** (scans your keymap for collisions), and **Safe Disable** which tears down overlays and handlers cleanly before you disable or upgrade the addon.

Upgrading? Run Safe Disable first, replace the zip, restart Bforartists. The addon also purges zombie classes on registration, which makes hot reinstalls reliable.

## 7. Layers — animation layer stack (Phase 11)

Stack-based animation layers with per-layer influence and blending, bone/part assignment operators, layer solo/mute workflows, and merge/bake-down paths. Layers integrate with the lipsync system (P12 lipsync layers are links in this stack).

## 8. Lipsync — PREVIEW/SHIPPED hybrid (Phase 12)

Phase 12 brings v12's headline feature: every lipsync layer link works in one of two modes. **PREVIEW** installs live drivers on your shape keys that read the cue table per frame — scrubbing responds instantly and no f-curves are written, ideal for blocking. **SHIPPED** removes the drivers and bakes the cue table into real shape-key f-curves — render-ready, NLA-safe, and exportable to engines without driver support.

One button — **Toggle Live Preview ↔ Shipped Bake** — performs the whole transition both ways (driver install/removal, key clear/bake). Targets: shape keys, bones, or both. Wiring is largely automatic: pick the mesh, then **Autofill Shape Key Wiring** name-matches visemes to shape-key blocks; capture custom viseme shapes when names don't match.

Hand-polish survives: the **manual override sanctuary** protects keys you edited (by frame) through rebakes, for shape keys exactly as for bones. The audio waveform shows in the Dope Sheet for sync reference.

Render warning: starting a render in PREVIEW mode triggers a console/UI warning — toggle to SHIPPED first so the render uses baked keys, not drivers. If the mouth freezes after reopening a file, toggle modes once (drivers re-register their namespace via a load handler).

## 9. Preferences & customization

Preferences include explainer-help visibility (full / icon-only / off), compact UI mode, performance mode, debug mode, drag sensitivity and autokey margin behavior for the offset tools, and fast-offset step size. Settings can be exported/imported as JSON for studio distribution (the only file permission the addon requests). A capabilities panel reports which subsystems initialized, and diagnostics panels expose self-checks.

## 10. Troubleshooting

| Symptom                                 | Resolution                                                                                                                                                |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Addon won't enable                      | Standard Blender is rejected by design (Bforartists-exclusive). On BFA, check the console — registration names the failing module and rolls back cleanly. |
| Viewport stutters with overlay on       | You're likely on pre-12.0.1 settings: set Max Samples ≈160, Max Display Targets ≤3, disable Constraint Evaluation, leave frame ticks off.                 |
| Arc diagnostics show nothing            | Since 12.0.1 detectors run only from the Run Diagnostics button (Motion tab or editor header), not automatically.                                         |
| Lipsync mouth frozen after opening file | PREVIEW drivers lost their namespace — toggle PREVIEW/SHIPPED once, or disable/enable the addon.                                                          |
| Render has no/wrong lipsync             | You rendered in PREVIEW. Toggle to SHIPPED before rendering (a render-time warning also fires).                                                           |
| Hotkeys conflict with other addons      | Workspace → Check Hotkey Conflicts lists collisions; pies default to Ctrl+Shift+D / Ctrl+Shift+K.                                                         |
| Errors while disabling/upgrading        | Use Workspace → Safe Disable first, then disable/replace the zip and restart.                                                                             |
| A panel is missing                      | Check the capabilities/diagnostics panel (Workspace) — a subsystem may have failed to initialize; the console identifies it.                              |

# Appendix A — Complete operator index

Every operator in v12.0.1, grouped by phase. Use F3 search or scripting (`bpy.ops.<id>`); each also has an in-app explainer entry where applicable. Full descriptions live in `AnimAssist_LLM_Reference.md`.

## Phase 2 — Keys (45)

| Identifier                                        | Label                                 |
|---------------------------------------------------|---------------------------------------|
| <code>animassist.clear_all_key_metadata</code>    | Clear All Key Metadata                |
| <code>animassist.clear_tag_selected_keys</code>   | Clear Tags on Selected Keys           |
| <code>animassist.copy_selected_keys</code>        | Copy Selected Keys                    |
| <code>animassist.deselect_all_visible</code>      | Deselect All Visible Keys             |
| <code>animassist.invert_channel_visibility</code> | Invert Channel Visibility             |
| <code>animassist.invert_selection</code>          | Invert Key Selection                  |
| <code>animassist.isolate_by_regex</code>          | Isolate Channels by Regex             |
| <code>animassist.isolate_custom_props</code>      | Isolate Custom Properties             |
| <code>animassist.isolate_selected_bones</code>    | Isolate Selected Bones                |
| <code>animassist.isolate_selected_channels</code> | Isolate Selected Channels             |
| <code>animassist.isolate_transform</code>         | Isolate Transform Channels            |
| <code>animassist.mirror_selected_keys</code>      | Mirror Selected Keys at Current Frame |
| <code>animassist.mute_unselected_channels</code>  | Mute Unselected Channels              |
| <code>animassist.offset_selected_frames</code>    | Offset Selected Keys (Frame)          |
| <code>animassist.offset_selected_values</code>    | Offset Selected Keys (Value)          |
| <code>animassist.paste_keys_at_frame</code>       | Paste Keys at Frame                   |
| <code>animassist.pop_channel_isolation</code>     | Restore Channel Isolation State       |
| <code>animassist.protect_selected_keys</code>     | Protect Selected Keys                 |
| <code>animassist.prune_orphan_key_metadata</code> | Prune Orphan Key Metadata             |
| <code>animassist.push_channel_isolation</code>    | Save Channel Isolation State          |
| <code>animassist.safe_delete_selected_keys</code> | Safe Delete Selected Keys             |
| <code>animassist.scan_dense_keys</code>           | Scan Dense Keys                       |
| <code>animassist.scan_redundant_keys</code>       | Scan Redundant Keys                   |
| <code>animassist.scan_spike_keys</code>           | Scan Spike Keys                       |
| <code>animassist.select_all_visible</code>        | Select All Visible Keys               |
| <code>animassist.select_between_selected</code>   | Select Between First/Last Selected    |
| <code>animassist.select_by_handle_type_adv</code> | Select Keys by Handle Type            |
| <code>animassist.select_by_interpolation</code>   | Select Keys by Interpolation          |
| <code>animassist.select_by_key_type_adv</code>    | Select Keys by Type                   |
| <code>animassist.select_by_tag</code>             | Select Keys by Tag                    |
| <code>animassist.select_by_value_range</code>     | Select Keys by Value Range            |
| <code>animassist.select_every_nth</code>          | Select Every Nth Key                  |

| Identifier                                          | Label                                |
|-----------------------------------------------------|--------------------------------------|
| <code>animassist.select_first_last</code>           | Select First/Last Keys               |
| <code>animassist.select_flat_segments</code>        | Select Keys in Flat Segments         |
| <code>animassist.select_frame_range</code>          | Select Keys in Frame Range           |
| <code>animassist.select_local_extremes</code>       | Select Local Extremes                |
| <code>animassist.select_neighbors</code>            | Extend Selection to Neighbours       |
| <code>animassist.select_playback_range</code>       | Select Keys in Playback Range        |
| <code>animassist.select_preview_range</code>        | Select Keys in Preview Range         |
| <code>animassist.select_protected</code>            | Select Protected Keys                |
| <code>animassist.set_key_flavor</code>              | Set Key Flavor                       |
| <code>animassist.set_key_note</code>                | Set Note on Selected Keys            |
| <code>animassist.show_all_channels</code>           | Show All Channels                    |
| <code>animassist.snap_keys_to_integer_frames</code> | Snap Selected Keys to Integer Frames |
| <code>animassist.tag_selected_keys</code>           | Tag Selected Keys                    |

## Phase 3 — Breakdowns (36)

| Identifier                                          | Label                      |
|-----------------------------------------------------|----------------------------|
| <code>animassist.apply_preset</code>                | Apply Preset               |
| <code>animassist.blend_toward_reference</code>      | Blend Toward Reference     |
| <code>animassist.breakdown_batch_frames</code>      | Batch Over Selected Frames |
| <code>animassist.breakdown_channel_subset</code>    | Custom Channel Subset      |
| <code>animassist.breakdown_current_frame</code>     | Breakdown at Current Frame |
| <code>animassist.breakdown_favor_next</code>        | Favor Next Pose            |
| <code>animassist.breakdown_favor_prev</code>        | Favor Previous Pose        |
| <code>animassist.breakdown_from_clipboard</code>    | Breakdown From Clipboard   |
| <code>animassist.breakdown_location_only</code>     | Location Only              |
| <code>animassist.breakdown_midpoint</code>          | Midpoint Breakdown         |
| <code>animassist.breakdown_numeric</code>           | Numeric Breakdown          |
| <code>animassist.breakdown_offset</code>            | Relative Offset            |
| <code>animassist.breakdown_percentage</code>        | Quick Percentage Breakdown |
| <code>animassist.breakdown_pull_next</code>         | Pull To Next               |
| <code>animassist.breakdown_pull_prev</code>         | Pull To Previous           |
| <code>animassist.breakdown_push_next</code>         | Push Into Next             |
| <code>animassist.breakdown_push_prev</code>         | Push From Previous         |
| <code>animassist.breakdown_repeat_last</code>       | Repeat Last Breakdown      |
| <code>animassist.breakdown_rotation_only</code>     | Rotation Only              |
| <code>animassist.breakdown_scale_only</code>        | Scale Only                 |
| <code>animassist.breakdown_selected_controls</code> | Selected Controls Only     |
| <code>animassist.breakdown_transform_only</code>    | Transform Only             |

| Identifier                         | Label                        |
|------------------------------------|------------------------------|
| animassist.breakdown_weighted      | Weighted Previous/Next       |
| animassist.commit_preview          | Commit Preview               |
| animassist.delete_preset           | Delete Preset                |
| animassist.inbetween_distribute    | Evenly Distribute Inbetweens |
| animassist.inbetween_on_clusters   | Inbetween on Clusters        |
| animassist.inbetween_selected_gap  | Inbetween in Selected Gap    |
| animassist.manage_exclusion_set    | Manage Exclusion Set         |
| animassist.modal_drag_breakdown    | Modal Drag Breakdown         |
| animassist.pose_compare_report     | Pose Compare Report          |
| animassist.pose_snapshot_next      | Set Next Pose                |
| animassist.pose_snapshot_prev      | Set Previous Pose            |
| animassist.pose_snapshot_reference | Set Reference Pose           |
| animassist.preview_breakdown       | Preview Breakdown            |
| animassist.save_preset             | Save Preset                  |

## Phase 4 — Offset (19)

| Identifier                          | Label                |
|-------------------------------------|----------------------|
| animassist.p4_apply_preset          | Apply Preset         |
| animassist.p4_invert_last           | Invert Last Offset   |
| animassist.p4_modal_offset          | Modal Drag Offset    |
| animassist.p4_nudge_current         | Nudge Current Frame  |
| animassist.p4_offset_rotate_only    | Rotation Only        |
| animassist.p4_offset_scale_only     | Scale Only           |
| animassist.p4_offset_selected       | Offset Selected Keys |
| animassist.p4_offset_translate_only | Translation Only     |
| animassist.p4_offset_trs_combined   | Combined TRS Offset  |
| animassist.p4_pull_x                | Pull X               |
| animassist.p4_pull_y                | Pull Y               |
| animassist.p4_pull_z                | Pull Z               |
| animassist.p4_push_x                | Push X               |
| animassist.p4_push_y                | Push Y               |
| animassist.p4_push_z                | Push Z               |
| animassist.p4_reapply_last          | Reapply Last Offset  |
| animassist.p4_set_pivot_cursor      | Pivot from 3D Cursor |
| animassist.p4_set_pivot_head        | Pivot from Bone Head |
| animassist.p4_set_pivot_tail        | Pivot from Bone Tail |

## Phase 5 — Trajectories (10)

| Identifier                        | Label                      |
|-----------------------------------|----------------------------|
| animassist.p5_disable_overlay     | Disable Trajectory Overlay |
| animassist.p5_enable_overlay      | Enable Trajectory Overlay  |
| animassist.p5_isolate_target      | Isolate Target             |
| animassist.p5_jump_next_issue     | Next Issue                 |
| animassist.p5_jump_prev_issue     | Previous Issue             |
| animassist.p5_mute_unselected     | Mute Unselected            |
| animassist.p5_refresh_overlay     | Refresh Trajectory Overlay |
| animassist.p5_run_diagnostics     | Run Arc Diagnostics        |
| animassist.p5_select_bad_arc_keys | Select Bad-Arc Keys        |
| animassist.p5_suggest_candidates  | Suggest Candidate Keys     |

## Phase 6 — Retiming (38)

| Identifier                      | Label                    |
|---------------------------------|--------------------------|
| animassist.p6_bake_timing       | Bake to Integer Frames   |
| animassist.p6_clear_diagnostics | Clear Diagnostics        |
| animassist.p6_clear_doubles     | Remove Duplicates        |
| animassist.p6_clear_range       | Clear Range              |
| animassist.p6_collapse_gap      | Collapse Gap at Playhead |
| animassist.p6_compress_timing   | Compress Timing          |
| animassist.p6_copy_diag_report  | Copy Report              |
| animassist.p6_detect_and_fill   | Detect & Fill Gaps       |
| animassist.p6_detect_gaps       | Detect Gaps              |
| animassist.p6_distribute_keys   | Distribute Keys          |
| animassist.p6_fill_gaps         | Fill Gaps                |
| animassist.p6_insert_time       | Insert Time              |
| animassist.p6_jump_next_cluster | Next Cluster             |
| animassist.p6_jump_next_gap     | Next Gap                 |
| animassist.p6_jump_prev_gap     | Previous Gap             |
| animassist.p6_match_timing      | Match Timing             |
| animassist.p6_modal_offset      | Interactive Offset       |
| animassist.p6_modal_scale       | Interactive Scale        |
| animassist.p6_normalize_spacing | Normalize Spacing        |
| animassist.p6_offset_keys       | Offset Keys              |
| animassist.p6_offset_range      | Offset Range             |
| animassist.p6_remove_time       | Remove Time              |
| animassist.p6_reset_timing      | Reset Timing             |
| animassist.p6_restore_range     | Restore Range            |
| animassist.p6_reverse_keys      | Reverse Keys             |



| Identifier                         | Label                   |
|------------------------------------|-------------------------|
| animassist.p6_ripple_backward      | Ripple Backward         |
| animassist.p6_ripple_forward       | Ripple Forward          |
| animassist.p6_ripple_to_end        | Ripple to End           |
| animassist.p6_run_diagnostics      | Run Diagnostics         |
| animassist.p6_scale_keys           | Scale Keys              |
| animassist.p6_scale_range          | Scale Range             |
| animassist.p6_select_keys_in_range | Select Keys in Range    |
| animassist.p6_set_pivot            | Set Pivot from Playhead |
| animassist.p6_set_range_end        | Set Range End           |
| animassist.p6_set_range_start      | Set Range Start         |
| animassist.p6_snap_to_frames       | Snap to Frames          |
| animassist.p6_store_range          | Store Range             |
| animassist.p6_time_warp            | Time Warp %             |

## Phase 7 — Proxies (48)

| Identifier                                | Label                      |
|-------------------------------------------|----------------------------|
| animassist.p7_apply_offset                | Apply Offset               |
| animassist.p7_auto_cleanup                | Auto Cleanup               |
| animassist.p7_bake_locator_from_target    | Bake Locator from Target   |
| animassist.p7_bake_preserve_timing        | Bake Preserve Timing       |
| animassist.p7_bake_preview                | Bake Preview Range         |
| animassist.p7_bake_range                  | Bake Range                 |
| animassist.p7_bake_selected               | Bake Selected              |
| animassist.p7_bake_selected_channels      | Bake Selected Channels     |
| animassist.p7_bake_step                   | Bake with Step             |
| animassist.p7_bake_target_from_locator    | Bake Target from Locator   |
| animassist.p7_batch_create_proxies        | Batch Create Proxies       |
| animassist.p7_check_setup                 | Check Setup                |
| animassist.p7_cleanup_all                 | Cleanup All Sessions       |
| animassist.p7_cleanup_session             | Cleanup Session            |
| animassist.p7_constrain_locator_to_target | Constrain Locator → Target |
| animassist.p7_constrain_target_to_locator | Constrain Target → Locator |
| animassist.p7_create_locator              | Create Locator at Target   |
| animassist.p7_create_locator_average      | Locator at Average         |
| animassist.p7_create_locator_cursor       | Locator at Cursor          |
| animassist.p7_create_proxy                | Create Proxy               |
| animassist.p7_export_session              | Export Session             |
| animassist.p7_list_sessions               | List Sessions              |

| Identifier                            | Label                      |
|---------------------------------------|----------------------------|
| animassist.p7_lock_target             | Lock Target                |
| animassist.p7_match_locator_to_target | Match Locator → Target     |
| animassist.p7_match_target_to_locator | Match Target → Locator     |
| animassist.p7_mirror_proxy            | Mirror Proxy               |
| animassist.p7_mute_constraints        | Mute Constraints           |
| animassist.p7_one_click_cleanup       | One-Click Cleanup          |
| animassist.p7_one_click_proxy_bake    | One-Click Proxy & Bake     |
| animassist.p7_parent_locator          | Parent Locator to Target   |
| animassist.p7_purge_artifacts         | Purge P7 Artifacts         |
| animassist.p7_quick_proxy             | Quick Proxy                |
| animassist.p7_recenter_proxy          | Recenter Proxy             |
| animassist.p7_reconnect_session       | Reconnect Session          |
| animassist.p7_recover_session         | Recover P7 Session         |
| animassist.p7_reduce_keys             | Reduce Keys                |
| animassist.p7_remove_constraints      | Remove Session Constraints |
| animassist.p7_remove_proxy            | Remove Proxy               |
| animassist.p7_rename_proxy            | Rename Proxy               |
| animassist.p7_set_proxy_color         | Set Proxy Color            |
| animassist.p7_show_session_info       | Session Info               |
| animassist.p7_smart_bake              | Smart Bake                 |
| animassist.p7_switch_proxy_mode       | Switch Proxy Mode          |
| animassist.p7_temp_pivot              | Temp Pivot                 |
| animassist.p7_toggle_collection       | Toggle Collection          |
| animassist.p7_toggle_display          | Toggle Proxy Display       |
| animassist.p7_validate_session        | Validate Session           |
| animassist.p7_zero_proxy              | Zero Proxy                 |

## Phase 8 — IK/FK (53)

| Identifier                           | Label                  |
|--------------------------------------|------------------------|
| animassist.p8_apply_detected_pattern | Apply Detected Pattern |
| animassist.p8_bake_switch_preview    | Bake Switch Preview    |
| animassist.p8_bake_switch_range      | Bake Switch Range      |
| animassist.p8_batch_switch           | Batch Switch           |
| animassist.p8_chain_nav              | Navigate Chains        |
| animassist.p8_chain_summary          | Chain Summary          |
| animassist.p8_chain_to_match         | Chain → Match          |
| animassist.p8_clear_history          | Clear Switch History   |
| animassist.p8_compensate_multi       | Compensate Multi       |

| Identifier                                | Label                       |
|-------------------------------------------|-----------------------------|
| animassist.p8_compensate_single           | Compensate Single           |
| animassist.p8_compensation_report         | Compensation Report         |
| animassist.p8_contact_mask_from_selection | Contact Mask from Selection |
| animassist.p8_contact_preserve_match      | Contact Preserve Match      |
| animassist.p8_debug_diagnostics           | Debug Diagnostics           |
| animassist.p8_delete_switch_preset        | Delete Switch Preset        |
| animassist.p8_detect_all                  | Detect All Patterns         |
| animassist.p8_detect_chain_for_bone       | Detect Chain for Bone       |
| animassist.p8_detect_chains               | Detect IK Chains            |
| animassist.p8_detect_custom_props         | Detect Custom Properties    |
| animassist.p8_detect_influence_patterns   | Detect Influence Patterns   |
| animassist.p8_detect_parent_patterns      | Detect Parent Patterns      |
| animassist.p8_detect_space_enums          | Detect Space Enums          |
| animassist.p8_find_bone_chains            | Find Chains for Bone        |
| animassist.p8_invalidate_chain_cache      | Clear Chain Cache           |
| animassist.p8_load_switch_preset          | Load Switch Preset          |
| animassist.p8_match_axis_filtered         | Match Axis Filtered         |
| animassist.p8_match_local_matrix          | Match Local Matrix          |
| animassist.p8_match_location              | Match Location              |
| animassist.p8_match_opposite              | Match Opposite              |
| animassist.p8_match_rotation              | Match Rotation              |
| animassist.p8_match_scale                 | Match Scale                 |
| animassist.p8_match_selected_to_active    | Match Selected to Active    |
| animassist.p8_match_to_parent             | Match to Parent             |
| animassist.p8_match_to_target             | Match to Target             |
| animassist.p8_match_to_world              | Match to World              |
| animassist.p8_match_trs                   | Match TRS                   |
| animassist.p8_match_visual_matrix         | Match Visual Matrix         |
| animassist.p8_match_with_offset           | Match with Offset           |
| animassist.p8_match_without_offset        | Match without Offset        |
| animassist.p8_nav_next_switch             | Next Switch                 |
| animassist.p8_nav_prev_switch             | Prev Switch                 |
| animassist.p8_quick_match                 | Quick Match                 |
| animassist.p8_repeat_last_switch          | Repeat Last Switch          |
| animassist.p8_restore_switch              | Restore Switch              |
| animassist.p8_save_switch_preset          | Save Switch Preset          |
| animassist.p8_select_chain_bones          | Select Chain Bones          |
| animassist.p8_switch_bool                 | Switch Bool                 |

| Identifier                        | Label                     |
|-----------------------------------|---------------------------|
| animassist.p8_switch_enum         | Switch Enum               |
| animassist.p8_switch_influence    | Switch Influence          |
| animassist.p8_switch_marker       | Switch Marker             |
| animassist.p8_toggle_preview      | Toggle Preview            |
| animassist.p8_unsupported_warning | Unsupported Setup Warning |
| animassist.p8_visual_match        | Visual Match              |

## Phase 9 — Mirroring (43)

| Identifier                             | Label                     |
|----------------------------------------|---------------------------|
| animassist.p9_add_naming_exception     | Add Naming Exception      |
| animassist.p9_add_opposite             | Add Opposite to Selection |
| animassist.p9_add_pair_override        | Add Pair Override         |
| animassist.p9_ambiguous_warning        | Ambiguous Pair Warning    |
| animassist.p9_batch_mirror             | Batch Mirror Selected     |
| animassist.p9_batch_mirror_active_side | Mirror Active Side        |
| animassist.p9_build_cache              | Build Pair Cache          |
| animassist.p9_channel_resolver         | Channel Resolver          |
| animassist.p9_custom_pattern           | Apply Custom Pattern      |
| animassist.p9_load_pair_preset         | Load Pair Preset          |
| animassist.p9_match_opposite_to_active | Match Opposite to Active  |
| animassist.p9_match_to_opposite        | Match to Opposite         |
| animassist.p9_mirror_frame             | Mirror Frame              |
| animassist.p9_mirror_keyed_only        | Mirror Keyed Only         |
| animassist.p9_mirror_local             | Mirror Local              |
| animassist.p9_mirror_location          | Mirror Location           |
| animassist.p9_mirror_metadata          | Mirror Metadata           |
| animassist.p9_mirror_pose              | Mirror Pose               |
| animassist.p9_mirror_preset_values     | Mirror Preset Values      |
| animassist.p9_mirror_preview           | Mirror Preview            |
| animassist.p9_mirror_proxy_helpers     | Mirror Proxy Helpers      |
| animassist.p9_mirror_range             | Mirror Range              |
| animassist.p9_mirror_report            | Mirror Report             |
| animassist.p9_mirror_rotation          | Mirror Rotation           |
| animassist.p9_mirror_scale             | Mirror Scale              |
| animassist.p9_mirror_selected          | Mirror Selected           |
| animassist.p9_mirror_selection_sets    | Mirror Selection Sets     |
| animassist.p9_mirror_switch_targets    | Mirror Switch Targets     |
| animassist.p9_mirror_visible_only      | Mirror Visible Only       |

| Identifier                            | Label                    |
|---------------------------------------|--------------------------|
| animassist.p9_mirror_with_offset      | Mirror with Offset       |
| animassist.p9_mirror_without_offset   | Mirror without Offset    |
| animassist.p9_mirror_world            | Mirror World             |
| animassist.p9_missing_warning         | Missing Opposite Warning |
| animassist.p9_nav_next_unpaired       | Next Unpaired            |
| animassist.p9_remove_naming_exception | Remove Naming Exception  |
| animassist.p9_remove_pair_override    | Remove Pair Override     |
| animassist.p9_repeat_mirror           | Repeat Last Mirror       |
| animassist.p9_save_pair_preset        | Save Pair Preset         |
| animassist.p9_select_opposite         | Select Opposite          |
| animassist.p9_swap_poses              | Swap Poses               |
| animassist.p9_swap_selection          | Swap Selection           |
| animassist.p9_validate_pairs          | Validate Pairs           |
| animassist.p9_visual_mirror           | Visual Mirror            |

## Phase 10 — Workspace (55)

| Identifier                            | Label                    |
|---------------------------------------|--------------------------|
| animassist.p10_add_favorite           | Add to Favorites         |
| animassist.p10_add_macro              | Add Macro                |
| animassist.p10_add_macro_step         | Add Macro Step           |
| animassist.p10_batch_bookmarked       | Batch: Bookmarked Frames |
| animassist.p10_batch_frame_steps      | Batch: Frame Steps       |
| animassist.p10_batch_selected         | Batch: Selected Targets  |
| animassist.p10_check_hotkey_conflicts | Check Hotkey Conflicts   |
| animassist.p10_clear_errors           | Clear Errors             |
| animassist.p10_clear_history          | Clear History            |
| animassist.p10_clear_recents          | Clear Recents            |
| animassist.p10_clear_snapshots        | Clear Snapshots          |
| animassist.p10_export_workspace       | Export Workspace Profile |
| animassist.p10_final_validation       | Final Validation         |
| animassist.p10_first_run_setup        | First Run Setup          |
| animassist.p10_import_workspace       | Import Workspace Profile |
| animassist.p10_leak_check             | Leak Check               |
| animassist.p10_list_snapshots         | List Snapshots           |
| animassist.p10_load_demo_config       | Load Demo Config         |
| animassist.p10_load_profile           | Load Profile             |
| animassist.p10_macro_breakdown_offset | Breakdown + Offset       |
| animassist.p10_macro_diagnose_jump    | Diagnose + Jump          |

| Identifier                             | Label                  |
|----------------------------------------|------------------------|
| animassist.p10_macro_mirror_match      | Mirror + Match         |
| animassist.p10_macro_proxy_workflow    | Proxy Workflow         |
| animassist.p10_macro_switch_compensate | Switch + Compensate    |
| animassist.p10_metadata_cleanup        | Metadata Cleanup       |
| animassist.p10_pie_breakdown           | Breakdown Pie          |
| animassist.p10_pie_key_tools           | Key Tools Pie          |
| animassist.p10_pie_proxy               | Proxy Controls Pie     |
| animassist.p10_pie_switch              | Space Switch Pie       |
| animassist.p10_pie_symmetry            | Symmetry Pie           |
| animassist.p10_pie_transform           | Transform Pie          |
| animassist.p10_rebuild_caches          | Rebuild Caches         |
| animassist.p10_record_recent           | Record Recent Tool     |
| animassist.p10_remove_favorite         | Remove Favorite        |
| animassist.p10_remove_macro            | Remove Macro           |
| animassist.p10_remove_macro_step       | Remove Macro Step      |
| animassist.p10_remove_profile          | Remove Profile         |
| animassist.p10_repeat_last             | Repeat Last Tool       |
| animassist.p10_reset_ui                | Reset UI State         |
| animassist.p10_restore_snapshot        | Restore Snapshot       |
| animassist.p10_run_custom_macro        | Run Custom Macro       |
| animassist.p10_safe_disable            | Safe Disable           |
| animassist.p10_save_profile            | Save Profile           |
| animassist.p10_search_tools            | Search Tools           |
| animassist.p10_show_errors             | Show Error Log         |
| animassist.p10_show_history            | Show Operation History |
| animassist.p10_show_stats              | Session Statistics     |
| animassist.p10_stale_cleanup           | Stale Cleanup          |
| animassist.p10_system_diagnostics      | System Diagnostics     |
| animassist.p10_tag_preset              | Tag Preset             |
| animassist.p10_take_snapshot           | Take Snapshot          |
| animassist.p10_toggle_debug            | Toggle Debug Panel     |
| animassist.p10_toggle_shelf_mode       | Toggle Shelf Mode      |
| animassist.p10_validate_macro          | Validate Macro         |
| animassist.p10_validate_registration   | Validate Registration  |

## Phase 11 — Layers (31)

| Identifier                          | Label                |
|-------------------------------------|----------------------|
| animassist.p11_add_channel_override | Add Channel Override |

| Identifier                             | Label                   |
|----------------------------------------|-------------------------|
| animassist.p11_add_layer               | Add Layer               |
| animassist.p11_assign_by_pattern       | Assign by Pattern       |
| animassist.p11_assign_selected         | Assign Selected         |
| animassist.p11_auto_partition          | Auto-Partition Layers   |
| animassist.p11_blend_layers            | Blend Between Layers    |
| animassist.p11_clear_assignments       | Clear Assignments       |
| animassist.p11_duplicate_layer         | Duplicate Layer         |
| animassist.p11_evaluate_stack          | Evaluate Layer Stack    |
| animassist.p11_flatten_all             | Flatten All Layers      |
| animassist.p11_init_layers             | Initialize Layers       |
| animassist.p11_load_preset             | Load Layer Preset       |
| animassist.p11_merge_down              | Merge Down              |
| animassist.p11_move_layer_down         | Move Layer Down         |
| animassist.p11_move_layer_up           | Move Layer Up           |
| animassist.p11_remove_bone             | Remove Bone             |
| animassist.p11_remove_channel_override | Remove Channel Override |
| animassist.p11_remove_layer            | Remove Layer            |
| animassist.p11_remove_preset           | Remove Preset           |
| animassist.p11_remove_selected         | Remove Selected         |
| animassist.p11_rename_layer            | Rename Layer            |
| animassist.p11_save_preset             | Save Layer Preset       |
| animassist.p11_select_assigned         | Select Assigned Bones   |
| animassist.p11_select_layer            | Select Layer            |
| animassist.p11_set_active              | Set Active Layer        |
| animassist.p11_set_blend_factor        | Set Blend Factor        |
| animassist.p11_set_blend_mode          | Set Blend Mode          |
| animassist.p11_set_weight              | Set Layer Weight        |
| animassist.p11_toggle_lock             | Toggle Lock             |
| animassist.p11_toggle_mute             | Toggle Mute             |
| animassist.p11_toggle_solo             | Toggle Solo             |

## Phase 12 — Lipsync (15)

| Identifier                               | Label                       |
|------------------------------------------|-----------------------------|
| animassist.p12_autofill_shape_key_wiring | Autofill Shape Key Wiring   |
| animassist.p12_autofill_wiring           | Autofill Rig Wiring         |
| animassist.p12_bake_lipsync              | Bake Lipsync                |
| animassist.p12_capture_viseme            | Capture Viseme              |
| animassist.p12_capture_viseme_shape      | Capture Viseme (Shape Keys) |

| Identifier                        | Label                                |
|-----------------------------------|--------------------------------------|
| animassist.p12_clear_auto_keys    | Clear Auto Keys                      |
| animassist.p12_install_drivers    | Install Lipsync Drivers              |
| animassist.p12_mark_manual        | Mark as Manual Override              |
| animassist.p12_pick_mesh          | Use Active Mesh                      |
| animassist.p12_rebake             | Rebake Lipsync                       |
| animassist.p12_refresh_audio_hash | Refresh Audio Hash                   |
| animassist.p12_remove_drivers     | Remove Lipsync Drivers               |
| animassist.p12_reset_viseme       | Reset to Built-in                    |
| animassist.p12_setup_lipsync      | Setup Lipsync                        |
| animassist.p12_toggle_mode        | Toggle Live Preview <-> Shipped Bake |

## Core & cross-phase (30)

| Identifier                              | Label                                        |
|-----------------------------------------|----------------------------------------------|
| anim_assist.copy_diagnostics            | Copy Diagnostics                             |
| anim_assist.export_settings             | Export AnimAssist Settings                   |
| anim_assist.import_settings             | Import AnimAssist Settings                   |
| anim_assist.refresh_diagnostics         | Refresh Diagnostics                          |
| anim_assist.show_help_popup             | Anim Assist Help                             |
| animassist.add_bookmark                 | Add Frame Bookmark                           |
| animassist.anim_offset                  | Anim Offset                                  |
| animassist.anim_offset_range_from_scene | Range from Scene                             |
| animassist.anim_offset_set_range        | Set Offset Range                             |
| animassist.anim_offset_toggle_mask      | Toggle Mask                                  |
| animassist.batch_interpolation          | Batch Set Interpolation                      |
| animassist.blend_frame                  | Blend Frame                                  |
| animassist.blend_neighbor               | Blend to Neighbor                            |
| animassist.blend_offset                 | Blend Offset                                 |
| animassist.deep_uninstall               | Anim Assist: Deep Uninstall (Purge All Data) |
| animassist.delete_by_key_type           | Delete by Key Type                           |
| animassist.ease_to_ease                 | Ease To Ease                                 |
| animassist.insert_frames                | Insert Frames                                |
| animassist.move_keys                    | Move Keys                                    |
| animassist.purge_zombie_classes         | Anim Assist: Purge Zombie Classes            |
| animassist.push_pull                    | Push / Pull                                  |
| animassist.recall_bookmark              | Recall Bookmark                              |
| animassist.remove_bookmark              | Remove Frame Bookmark                        |
| animassist.report_saved_versions        | Anim Assist: Check Saved Schema Versions     |
| animassist.select_by_key_type           | Select by Key Type                           |



| Identifier                                  | Label               |
|---------------------------------------------|---------------------|
| <code>animassist.select_handles</code>      | Select Handles      |
| <code>animassist.set_handle_type</code>     | Set Handle Type     |
| <code>animassist.set_key_type</code>        | Set Key Type        |
| <code>animassist.set_reference_frame</code> | Set Reference Frame |
| <code>animassist.smooth_keys</code>         | Smooth Keys         |

Companion file for AI assistants: `AnimAssist_LLM_Reference.md` — full descriptions for all operators, 405 in-app help entries, architecture invariants and troubleshooting trees.